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Abstract. This is an example of ALEA's LaTeXstyle available at ALEA's website. Compile it preferentially using `pdflatex`.

1. Introduction

1.1. *Citations.*

- Compile bibliography using `bibtex` (to find bibliographical entries, MathSciNet database may be useful).
- Whenever it is possible, provide a DOI reference for the citation.
- In this latex style the `\cite` command creates citations using the name of the author and the date of publishing, for example Scott (1900).
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- It is very important to attach the file *.bib* with all references (see `example.bib`).

1.2. *Graphics.*

- To create graphics, include images preferentially in `Tikz` (see Figure 1.1) or png (see Figure 2.2), but other formats may be also accepted.

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- Please include all files of external figures in your final submission.

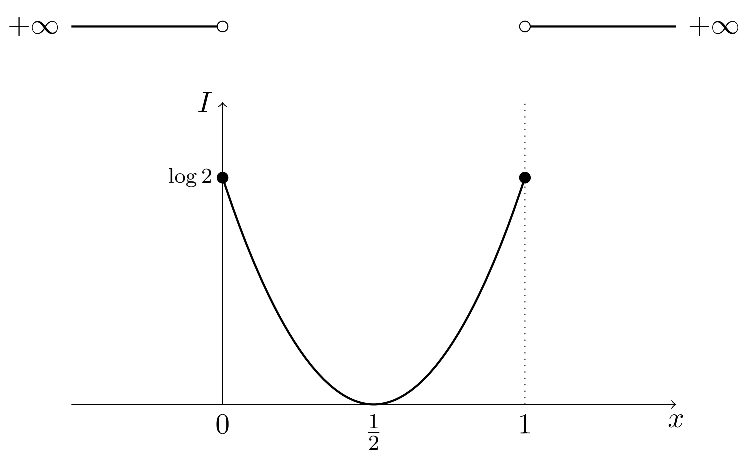

FIGURE 1.1. Rate function $I : \mathbb{R} \to [0, \infty]$

1.3. *Statements.*

- Tagged and untagged formulas are presented in the next section, where we further illustrate the use of the commands `\section`, `\subsection`, and environments as `\begin{theorem} ... \end{theorem}`.
- Theorems, lemmas, propositions etc. should be referred as Theorem 2.2 for instance, and sections as Section 2 for example.
- Make reference to equations using `\eqref{}` and make references to theorem, lemmas, etc., using `\ref{}`.
- **Avoid using** `\left...\right`. Prefer using predetermined sizes, such as `\big(`, `\Big(`, `\bigg(` and `\Bigg(`.
- **Do not use** the environment `\begin{eqnarray} ... \end{eqnarray}` and prefer instead the environments `\begin{align}...\end{align}` or `\begin{align*}...\end{align*}` or `\begin{multline} ... \end{multline}` or `\begin{multline*}...\end{multline*}`.

2. Dummy text example

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The infinitesimal generator $\big(A_L f\big)(n)$ is equal to

$$\frac{A}{L^a}\Big[f(n_{\delta(-)}+n_{-L},0,n_{-L+1},\dots,n_{\delta(+)})-f(n)\Big] \tag{2.1}$$
$$+\frac{B}{L^b}\Big[f(n_{\delta(-)},\dots,n_{L-1},0,n_{\delta(+)}+n_L)-f(n)\Big]$$
$$+\sum_{j=-L}^{L}\frac{1}{n_j+n_{j+1}+1}\sum_{q=0}^{n_j+n_{j+1}}\Big[f(n_{\delta(-)},n_{-L},\dots,n_{i-1},q,n_i+n_{i+1}-q,\dots,n_{\delta(+)})-f(n)\Big], \tag{2.2}$$

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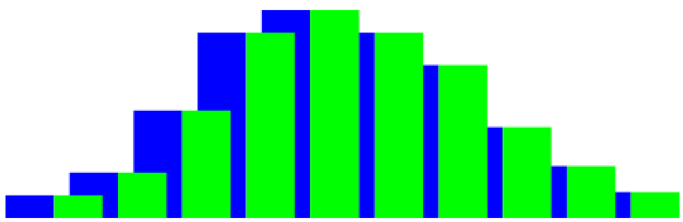

FIGURE 2.2. Some distribution

2.1. *Alea jacta est.* Fusce id eros vel lorem iaculis vulputate. Mauris imperdiet massa nulla, et dapibus tortor. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Mauris dictum consequat dui, ut consequat ligula convallis ut. Phasellus nisl magna, venenatis eget pharetra ac, commodo eu odio. Mauris porta pharetra metus. Nulla molestie porta erat, in tincidunt sem consequat ut. Morbi mollis facilisis imperdiet. Nunc vulputate, urna at rhoncus ullamcorper, lectus lectus facilisis massa, id condimentum est ipsum sed dolor. Sed accumsan ligula quis urna ornare ut euismod nisi venenatis. Morbi nulla nibh, tincidunt sit amet dignissim a, rhoncus a nibh.

Lemma 2.1. *Pellentesque nibh turpis, $\Omega \neq \varnothing$, Vivamus vitae massa a lorem eleifend dignissim eget eu orci. Nunc vel:*

$$\mathbb{P}(X_{n+1}=x\,|\,\mathcal{G}_n) \;=\; \frac{e^{\beta\,c_n(X_n,x)}}{\sum_{y\sim X_n} e^{\beta\,c_n(X_n,y)}}\,.$$

Proof: Sed a elit quis enim tincidunt dignissim vitae eget erat. Quisque tempus ante semper dolor scelerisque mollis. Nulla libero odio, vulputate non iaculis a, placerat at justo. Praesent luctus eleifend purus vel elementum. Nulla enim ligula, sagittis ut pretium a, scelerisque ut eros.

$$\begin{aligned}|F| &\lesssim \Big(\mathbb{P}_0\big[X_{s_2n^2}=-1\big]+\mathbb{P}_0\big[X_{s_2n^2}=0\big]\Big) \\ &\quad\times\left(\frac{1}{((s_1-s_2)n^2)^2}+\Big|K_{(s_1-s_2)n^2}(0)-K_{(s_1-s_2)n^2}(1)\Big|\right) \\ &\lesssim \frac{1}{\sqrt{s_2}}\cdot\frac{1}{(s_1-s_2)^{3/2}n^4}\,. \qquad (2.3)\end{aligned}$$

Sed auctor justo id elit volutpat bibendum. In aliquet dui vel mi consectetur dictum. Vestibulum vel lectus ut lacus porttitor bibendum sit amet nec erat (2.3). Fusce nec elit ante, sit amet ullamcorper nisl. Sed a tellus sed augue molestie accumsan.

$$\sum_{k=0}^{n}\binom{n}{k}a^k(1-a)^{n-k} \;=\; 1\,.$$

Praesent nisi erat, sollicitudin eu lacinia quis, tempus eu velit. Nulla magna erat, rhoncus eu suscipit et, rutrum vitae turpis. Vivamus sed neque leo. Donec id felis ut est viverra. □

Theorem 2.2. *Lorem ipsum dolor sit amet* $\gamma \in \mathbb{R}$, *consectetur adipiscing elit* $j \in \mathbb{Z}_+$, $\tau \in \mathbb{Z}$. *Pellentesque nibh turpis, sagittis ac scelerisque et, rhoncus et enim. Nunc vel:*

$$\lim_{n\to\infty}\frac{1}{a_n}\log\Big(\int_X e^{a_nF(x)}\,P_n(dx)\Big) \;=\; \sup_{x\in X}\Big\{F(x)-I(x)\Big\}. \qquad (2.4)$$

Proof: By Lemma 2.1, aenean consequat, urna in congue viverra, enim nunc suscipit lorem (2.4). □

Acknowledgements

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References

Finzi, B. *Meccanica razionale. Volume primo. Teorie introduttive, cinematica, statica.* Nicola Zanichelli Editore, Bologna (1950). Seconda edizione. MR0050410.

Huntington, E. V. and Whittemore, J. K. Some curious properties of conics touching the line infinity at one of the circular points. *Bull. Amer. Math. Soc.*, **8** (3), 122–124 (1901). MR1557852.

Scott, C. A. The International Congress of Mathematicians in Paris. *Bull. Amer. Math. Soc.*, **7** (2), 57–79 (1900). MR1557756.

Scott, C. A. Book Review: Compte rendu du deuxième Congrès international des Mathématiciens tenu à Paris du 6 au 12 août 1900. *Bull. Amer. Math. Soc.*, **9** (4), 214–215 (1903). MR1557978.